

The Learning Problem

“Probability is the very guide of life.”

—Joseph Butler, 1736

Suppose we are given a list of input–output pairs $(x_1, y_1), \dots, (x_N, y_N)$ and we want to produce a function f that, given a fresh x , returns a sensible guess for y . This task is called *supervised learning*, and almost everything we shall do in this course can be phrased as the search for such an f .

Before searching, we have to say what *sensible* means. The present lecture sets up the *functional risk*—the ideal quantity we would like to make small—and then exhibits the two compromises that turn its idealized minimum into something a computer can actually approximate.

1.1 FOUR INGREDIENTS

A supervised learning problem is specified by four objects.

1. A *data space* $\Omega = X \times Y$. The first factor is the *input space* (typically $X \subseteq \mathbb{R}^d$); the second is the *label space* (typically $Y \subseteq \mathbb{R}^p$, or a finite set of class indices).
2. A *model*, i.e., a function $f: X \rightarrow Y$. Which functions we are willing to consider becomes important in §1.4.
3. A *loss function* $\ell: Y \times Y \rightarrow \mathbb{R}_+$, measuring how close two labels are. The running example throughout these notes is the quadratic loss $\ell(a, b) = \frac{1}{2} \|a - b\|^2$.
4. A *probability measure* μ on Ω , so that (Ω, μ) is a probability space. This is the joint law of the pair (X, Y) from which all our data are drawn.

1.2 FUNCTIONAL RISK

Given the four ingredients of §1.1, the natural way to score a candidate model f is to average its pointwise loss against the joint law μ .

Definition (functional risk). The *functional risk* of a model f is

$$L(f) = \mathbb{E}(\ell \circ (f \times \text{id})) = \int_{\Omega} \ell(f(x), y) \, d\mu(x, y). \quad (1.1)$$

The *domain* of L is the set of measurable models for which the integral above is finite,

$$\text{dom } L = \{ f: X \rightarrow Y \text{ measurable} : L(f) < +\infty \}. \quad (1.2)$$

A running example

Take $\Omega = \mathbb{R}^2$ and let μ be the probability measure defined, for every Borel set $A \subseteq \mathbb{R}^2$, by

$$\mu(A) = \frac{1}{\pi} \int_A e^{-(x-y)^2 - x^2} \, dx \, dy, \quad (1.3)$$

the factor $1/\pi$ being chosen so that $\mu(\mathbb{R}^2) = 1$. Since μ is absolutely continuous with respect to the Lebesgue measure on $\mathbb{R}^2$, it is described by its *probability density function* $\rho := d\mu/d\lambda$,

$$\rho(x, y) = \frac{1}{\pi} e^{-(x-y)^2 - x^2}, \quad (1.4)$$

in the sense that $\mu(A) = \int_A \rho \, d\lambda$ for every Borel A .

The density factors as $\rho(x, y) = (e^{-x^2}/\sqrt{\pi}) \cdot (e^{-(x-y)^2}/\sqrt{\pi})$, a product of two Gaussians of variance $1/2$: the first peaked at $x = 0$, the second peaked (in y , with x fixed) at $y = x$. The mass of μ therefore concentrates around the line $y = x$ with Gaussian decay perpendicular to it (Figure 1.1). The *expected behaviour* of f is, accordingly, $f(x) = x$.

An equivalent picture: take a sample $(x, y) \sim \mu$, apply f to get $f(x)$, compare it to y via ℓ , and average over all samples weighted by μ .

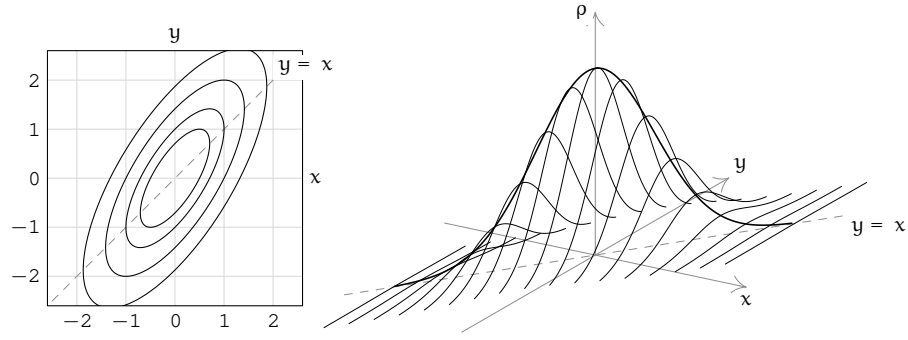


Figure 1.1. The joint density $\rho(x, y) = (1/\pi) e^{-(x-y)^2 - x^2}$. Left: level ellipses of the quadratic form $2x^2 - 2xy + y^2$; the dashed line $y = x$ is where ρ is maximized at each fixed x . Right: the same density as a 3D surface, sampled as y -slices at $x \in \{-2, -1.5, \dots, 2\}$; the ridge runs along $y = x$ with peak height decreasing as $|x|$ grows.

Computing the minimum

This example is one of the rare ones in which L can be minimized in closed form. We follow the *direct method*: find a candidate by stationarity, then verify a posteriori that the candidate really is a minimum.

Expanding the square in $\ell = \frac{1}{2}(f(x) - y)^2$ and pulling the prefactor in front,

$$L(f) = \frac{1}{2\pi} \int_{\mathbb{R}^2} (f^2(x) - 2f(x)y + y^2) e^{-(x-y)^2} e^{-x^2} dx dy. \quad (1.5)$$

The y -integral is handled by three Gaussian moments; substituting $z = y - x$ in each turns them into integrals of e^{-z^2} , $(z + x) e^{-z^2}$ and $(z + x)^2 e^{-z^2}$, so

$$\begin{aligned} \int_{\mathbb{R}} e^{-(x-y)^2} dy &= \sqrt{\pi}, \\ \int_{\mathbb{R}} y e^{-(x-y)^2} dy &= \sqrt{\pi} x, \\ \int_{\mathbb{R}} y^2 e^{-(x-y)^2} dy &= \sqrt{\pi} \left(x^2 + \frac{1}{2} \right), \end{aligned} \quad (1.6)$$

the last using $\int_{\mathbb{R}} z^2 e^{-z^2} dz = \sqrt{\pi}/2$. Substituting,

$$L(f) = \frac{1}{2\sqrt{\pi}} \int_{\mathbb{R}} \left(f^2(x) - 2xf(x) + x^2 + \frac{1}{2} \right) e^{-x^2} dx. \quad (1.7)$$

The terms in $x^2 + \frac{1}{2}$ do not depend on f , so minimizing L is equivalent to

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minimizing

$$J(f) := \int_{\mathbb{R}} (f^2(x) - 2x f(x)) e^{-x^2} dx. \quad (1.8)$$

Setting to zero the variation with respect to $f(x)$ pointwise gives the stationarity condition $2f(x) - 2x = 0$, i.e.

$$f^*(x) = x. \quad (1.9)$$

f^* is a minimum

Stationarity provided a candidate; we now prove that it really is a minimum, in two clean steps.

Step 1: $L(f) \geq 1/4$ for every $f \in \text{dom } L$. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be measurable (the defining requirement of $\text{dom } L$). Then $x \mapsto f(x) - x$ is measurable as a difference of two measurable functions, and $(f(x) - x)^2 e^{-x^2}$ is therefore a non-negative measurable function on $\mathbb{R}$. Now complete the square inside the integrand of (1.7):

$$f^2(x) - 2x f(x) + x^2 + \frac{1}{2} = (f(x) - x)^2 + \frac{1}{2}, \quad (1.10)$$

which turns (1.7) into

$$\begin{aligned} L(f) &= \frac{1}{2\sqrt{\pi}} \int_{\mathbb{R}} [(f(x) - x)^2 + \frac{1}{2}] e^{-x^2} dx \\ &= \frac{1}{2\sqrt{\pi}} \int_{\mathbb{R}} (f(x) - x)^2 e^{-x^2} dx + \frac{1}{4}, \end{aligned} \quad (1.11)$$

the last equality using $\int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\pi}$. The first summand is non-negative, so $L(f) \geq 1/4$.

Step 2: equality is attained at $f^*(x) := x$. Substituting f^* in the formula above kills the first summand, leaving $L(f^*) = 1/4$.

Combining the two steps,

$$\min\{L(f) : f \in \text{dom } L\} = \frac{1}{4}, \quad f^*(x) = x. \quad (1.12)$$

The minimizer is unique up to modification on a Lebesgue-null set: the first summand in Step 1 vanishes if and only if $f(x) = x$ for Lebesgue-a.e. $x \in \mathbb{R}$.

Two obstacles

Even when one is willing to take L as *the* quantity to minimize, two problems prevent us from doing so directly.

(P1) We do not know μ .

The constant 1/4 is the irreducible noise floor: it equals $\frac{1}{2} \cdot \frac{1}{2}$, the second factor being the integral $\int_{\mathbb{R}} (y - x)^2 e^{-(x-y)^2} / \sqrt{\pi} dy = 1/2$. No model, however clever, can do better.

(P2) Even if we did, minimizing over all measurable f is a variational problem we have no efficient algorithm to solve.

The next two sections describe the standard responses: replace μ by an empirical estimate, and replace “all measurable f ” by a parametric family.

1.3 EMPIRICAL RISK

We never see μ directly. What we see is a finite i.i.d. sample

$$\mathcal{D} = \{(x_1, y_1), \dots, (x_N, y_N)\}, \quad (x_i, y_i) \sim \mu. \quad (1.13)$$

The crudest possible substitute for μ is the *empirical measure*

$$\mu_N = \frac{1}{N} \sum_{i=1}^N \delta_{(x_i, y_i)}, \quad (1.14)$$

a sum of Dirac masses, one per data point. Replacing μ by μ_N inside L collapses the integral into a finite average:

$$L_N(f) = \frac{1}{N} \sum_{i=1}^N \ell(f(x_i), y_i). \quad (1.15)$$

This is the *empirical risk*. It depends only on quantities in hand, and the law of large numbers gives $L_N(f) \rightarrow L(f)$ pointwise in f as $N \rightarrow \infty$. Whether the *minimizer* of L_N tells us anything about the minimizer of L is a separate question; §1.5 will gesture at it.

1.4 RESTRICTING TO A HYPOTHESIS CLASS

The second compromise addresses (P2). Instead of minimizing L_N over the full space $\text{dom } L$, we choose a *hypothesis class* $\mathcal{M}$, typically a parametric family

$$\mathcal{M} = \{f(\cdot; w) : w \in \mathbb{R}^n\}, \quad (1.16)$$

indexed by a vector $w \in \mathbb{R}^n$ of *parameters*. The toy example, with $X = Y = \mathbb{R}$, is the family of affine functions:

$$f(x; a, b) = ax + b, \quad (a, b) \in \mathbb{R}^2. \quad (1.17)$$

Later in the course $\mathcal{M}$ will be a class of *neural networks*.

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The problem we actually solve is no longer variational but finite-dimensional. Setting $F(w) = L_N(f(\cdot; w))$,

$$f^N = f(\cdot; w^N), \quad w^N \in \arg \min \{ F(w) : w \in \mathbb{R}^n \}, \quad (1.18)$$

and the gradient descent of §2.1 can attack it directly.

The price of the two compromises. Write $L^* = \inf \{ L(f) : f \in \text{dom } L \}$ and $L_{\mathcal{M}}^* = \inf \{ L(f) : f \in \mathcal{M} \}$. The gap between the risk of our learned model f^N and the ideal risk L^* decomposes cleanly:

$$L(f^N) - L^* = \underbrace{L(f^N) - L_{\mathcal{M}}^*}_{\text{estimation error}} + \underbrace{L_{\mathcal{M}}^* - L^*}_{\text{approximation error}}. \quad (1.19)$$

The approximation error is the price for restricting to $\mathcal{M}$; it does not vanish even with infinitely many samples. The estimation error is the price for using L_N in place of L ; it should shrink as N grows. The whole game of statistical learning is to balance these two.

1.5 GENERALIZATION AND UNIFORM CONVERGENCE

[To be developed in a later lecture.]

For the estimation error $L(f^N) - L_{\mathcal{M}}^*$ to be small, it is not enough that $L_N(f) \rightarrow L(f)$ for each *fixed* f : we are choosing f^N as a function of the data, so the convergence must be *uniform* over the class $\mathcal{M}$. Quantifying “uniform” leads to capacity measures (VC dimension, Rademacher complexity, covering numbers) and to concentration inequalities; we shall come back to this once the needed tools are in hand.

1.6 A FIRST NUMERICAL EXPERIMENT

It is worth confirming the abstract claims of §1.3–1.4 with one’s own hands. Sampling from μ in the example of §1.2 is direct: generate two independent standard normals ξ_1, ξ_2 and set

$$x = \xi_1/\sqrt{2}, \quad y = x + \xi_2/\sqrt{2}. \quad (1.20)$$

Take the affine class $\mathcal{M}$ of §1.4 and write $f(x; w) = w_1 x + w_2$. The empirical risk and its gradient are

$$F(w) = L_N(f(\cdot; w)) = \frac{1}{2N} \sum_{i=1}^N (w_1 x_i + w_2 - y_i)^2, \quad (1.21)$$

$$\nabla F(w) = \frac{1}{N} \sum_{i=1}^N (w_1 x_i + w_2 - y_i) \begin{pmatrix} x_i \\ 1 \end{pmatrix}. \quad (1.22)$$

Run the gradient descent

$$w^{k+1} = w^k - \tau \nabla F(w^k), \quad w^0 \in \mathbb{R}^2, \quad (1.23)$$

to convergence; call the limit $w^N = (a^N, b^N)$ and the resulting model $f^N(x) = a^N x + b^N$.

The experiment is to repeat the whole procedure for a grid of sample sizes, say $N \in \{10, 10^2, 10^3, 10^4\}$, and to plot:

- (a) the parameter error $|a^N - 1| + |b^N|$, which should decay like $1/\sqrt{N}$;
- (b) the held-out risk $\tilde{L}_M(f^N) = (1/M) \sum_{j=1}^M \frac{1}{2} (f^N(x'_j) - y'_j)^2$ estimated on an independent test sample of size M , which should approach the analytic value $L^* = 1/4$;
- (c) the *training* risk $L_N(f^N)$, which approaches $1/4$ from *below*: it is biased downward by the very fact that f^N was chosen to minimize it.

Point (c) is the first concrete glimpse of why uniform convergence matters: L_N underestimates L precisely at its own minimizer, and quantifying that gap is the substance of §1.5.

Exercises

- 1 For the density of §1.2, compute $\int_{\mathbb{R}} \rho(x, y) dy$ and verify that the constant $C = 1/\pi$ is forced by $\int_{\mathbb{R}^2} \rho = 1$.
- 2 With $\mathcal{M} = \{f(x; a, b) = a x + b : (a, b) \in \mathbb{R}^2\}$ and the joint density of §1.2, compute $\arg \min \{L(f) : f \in \mathcal{M}\}$. Compare with f^* : is the approximation error zero?
- 3 Let L_N be the empirical risk built from an i.i.d. sample of size N drawn from μ , and let μ^N denote the product measure on Ω^N . Show that $\int_{\Omega^N} L_N(f) d\mu^N = L(f)$ for every fixed $f \in \text{dom } L$, and compute $\int_{\Omega^N} (L_N(f) - L(f))^2 d\mu^N$ as a function of N .

The least-squares minimizer is available in closed form: use it as a sanity check on the gradient-descent code.